One Checkpoint for Complete, Slide-Only and RNA-Only Patients: Auditing and Training Histology–Transcriptomics Survival Models for Missing Inputs

Sharoon Sharif and Ha Can

***Abstract*—Survival models that fuse whole-slide images with transcriptomics are evaluated on patients who have both inputs, whereas a hospital cohort also holds slide-only and RNA-only patients. Two questions are then open: does the fused model use both inputs, and what happens when one is missing? We audit this on five cohorts of The Cancer Genome Atlas under one equal-budget, three-seed protocol by re-evaluating trained checkpoints with an input removed or permuted across patients. Under this protocol, the official SurvPath implementation ranks patients the same way when its RNA input is permuted, with a change in concordance index (C-index) of $+0.001$, but loses 0.068 when the slide is permuted and falls to 0.47–0.52 without it. A query-bottleneck fusion model trained with learned null codes and modality dropout is equally invariant to RNA removal but loses only 0.030 without the slide. Adding auxiliary unimodal heads makes the fused prediction respond to RNA (-0.023 under permutation); the resulting checkpoint keeps a C-index of 0.54–0.62 with either input removed and moves at most 0.025 when half the patients lack a modality, so one trained model serves complete, slide-only and RNA-only patients. With both inputs it does not detectably beat an RNA-only multilayer perceptron or a late-fusion ensemble, and its $+0.036$ margin over the untuned SurvPath is significant under the Wilcoxon test but not the t-test. Replayed on stored training histories, validation-loss early stopping would have cost the two variants 0.06–0.09 C-index and SurvPath nothing. Code, per-run results and checkpoints are released.**

***Index Terms*—Computational pathology, missing modality, model auditing, multimodal fusion, survival analysis, transcriptomics, whole-slide images.**

I. Introduction

In routine care, tumor sequencing is ordered for a subset of patients and slide digitization is incomplete, so a prognostic cohort is rarely complete: some patients have a whole-slide image (WSI), some have bulk RNA-seq and some have both. Models that fuse the phenotype recorded by histology with the molecular state recorded by transcriptomics report higher concordance than either modality alone on most cohorts of The Cancer Genome Atlas (TCGA) [1]–[3]: pathology foundation models [4]–[6] turn a gigapixel WSI into a few thousand frozen patch embeddings, and co-attention between patch tokens and gene or pathway tokens [2], [7] relates the two sets. Such models are evaluated on patients who have both inputs, which leaves open two questions that an accuracy table cannot answer. Does the fused model use both inputs? A concordance index (C-index) measures how well predicted risks rank patients, not which input produced the ranking, so a model that silently ignores one input is a hidden failure. And what happens when one input is missing? A model that needs both cannot serve the incomplete patients.

We answer both questions with an input-use audit: every trained checkpoint is re-evaluated with one input removed, with one input permuted across patients and with a random fraction of patients lacking a modality, and its fused risks are correlated with those of its own single-input evaluations. Applied under one protocol to the official SurvPath implementation [2], the audit finds a ranking that does not respond to the RNA input (ΔC-index $+0.001$ under permutation across patients) and collapses when the slide is permuted (-0.068) or removed (C-index 0.47–0.52).

The practical result is a training recipe under which one trained model serves complete, slide-only and RNA-only patients: we train one query-bottleneck fusion model, NullQ-Former, in two variants that differ only in training. With learned null codes and modality dropout alone, the model is as invariant to RNA as SurvPath but no longer collapses without the slide. Adding auxiliary unimodal survival heads makes the fused prediction respond to RNA; the resulting checkpoint keeps a C-index of 0.54–0.62 with either input removed and changes by at most 0.025 when half of the patients lack a modality. With both inputs present, neither variant is detectably better than an RNA-only multilayer perceptron (MLP) or a late-fusion ensemble of two unimodal models, so the recipe buys coverage of incomplete patients, not accuracy. Our contributions are threefold. 1) A one-checkpoint training recipe assembled from existing components (null codes, modality dropout, auxiliary unimodal heads), with matched-variant evidence on where its missing-modality robustness comes from (Sections IV-C and IV-D). 2) An input-use audit (removal, permutation across patients, risk-score correlation; 3 seeds $\times$ 5 folds $\times$ 5 cohorts) of a published co-attention model under an equal-budget protocol (Section IV-A): SurvPath's ranking, as trained here, does not respond to its RNA input, so a high fused C-index does not show that both modalities

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S. Sharif is with the Department of Electrical Engineering and Computer Science, Alabama A&M University, Normal, AL 35762 USA, and with Faith Comes By Hearing, Albuquerque, NM USA (e-mail: sharoonsharif@gmail.com).

H. Can is an independent researcher (e-mail: canhaB139@gmail.com).

are used. 3) Protocol evidence that this literature rarely reports (Sections IV-B and IV-E): equal-budget accuracy with seed-aware statistics (seed-averaged 25-fold paired tests, bootstrap confidence intervals, Holm correction) and a post hoc replay of checkpoint selection on event-poor folds from the stored training histories.

Scope of claims. SurvPath was run with its published hyperparameters and was not re-tuned under our protocol. NullQ-Former's hyperparameters were developed on the bladder cohort (BLCA) under a 10-epoch budget, and the auxiliary-head variant and its loss weight were adopted after the 20-epoch results on all five cohorts had been seen, so both variants are reported as co-primary and no accuracy superiority over SurvPath is claimed. The training recipe has not been applied to SurvPath. All results are on TCGA with frozen UNI2-h features, simulated missingness is random and therefore uninformative, and no clinical covariates are modeled.

II. Related Work

Fusion of WSIs and genomics. WSIs are processed as bags of patches under multiple-instance learning (MIL): attention-based MIL (ABMIL) [8], CLAM [9] and TransMIL [10] pool patch features from self-supervised [11] or foundation-model encoders trained on 10^8 patches or more [4]–[6]. Early fusion work concatenated convolutional-network and genomic features [12] or combined unimodal representations with Kronecker products and gating [13]; a pan-cancer study found fusion to improve concordance in 12 of 14 cancer types [1]. MCAT [7] introduced co-attention from functional gene groups to patches; SurvPath [2] replaced the groups with Reactome and Hallmark pathway tokens and modeled dense pathway–patch interactions in one Transformer; MOTCat [14], CMTA [15], PIBD [16] and MMP [3], which reports gains over SurvPath, vary the interaction, and continually evolved multimodal foundation models extend the setting [17]. Two-bank learned-query fusion over patches and pathways already exists in APL [18]; the query bottleneck used here is not claimed as new.

Missing modalities in multimodal survival. Modality dropout was introduced for gesture recognition [19] and applied to pan-cancer prognosis by Cheerla and Gevaert [20] and MultiSurv [21]. Ma et al. [22] showed that multimodal Transformers are not robust to missing inputs by default; SMIL [23], ShaSpec [24] and missing-aware prompts [25] address it with meta-learning, shared–specific features or prompt tokens. For WSIs and genomics, masked-input training lets one model accept any subset of modalities [26], PathOmics [27] allows histology-only inference after multimodal training, G-HANet [28] distills genomic knowledge into a histology-only model, and recent work binds incomplete modalities [29], distills prompts [30], learns conditional latent variables [31], uses prototype-guided cross-modal knowledge [32], memory-augmented hypergraphs [33] or prototype-driven pretraining [34]. Permutation-based tests of which modality a model perceives exist for vision–language models [35]; none of the survival works above audits whether a published co-attention model uses its genomic input or tests under equal budgets whether unimodal auxiliary supervision changes that, which is the gap addressed here.

TABLE I
COHORTS AS REALIZED BY THE DSS SPLITS (THE SAME FIVE FOLDS FOR EVERY METHOD AND SEED); PER-FOLD COLUMNS: MINIMUM–MAXIMUM OVER THE FOLDS.

Cohort	Patients	DSS events	Patients/fold	Events/fold
BLCA†	359	113	70–75	16–25
BRCA	868	60	166–182	7–20
COADREAD	296	37	45–71	5–13
HNSC	392	117	56–126	13–45
STAD	318	84	42–101	9–33

†Development cohort for NullQ-Former's hyperparameters.

Learned queries and modality imbalance. Learned vectors that cross-attend to a large input and return a fixed-size summary (inducing points [36], object queries [37], Perceiver latents [38], [39], the Q-Former [40], the prototype banks of APL [18]) have the one property used here: the output has a fixed shape, so an absent input can be replaced by a learned tensor of the same shape. Joint training lets one modality dominate (modality competition [41]); Gradient Blending [42] and OGM-GE [43] counteract it with per-modality supervision or gradient modulation. The auxiliary unimodal heads used here are unimodal supervision in that tradition, not a new idea; our contribution is to measure what they change in a survival-fusion model under an audit.

III. Materials and Methods

A. Data and Cohorts

Five TCGA cohorts are used (Table I): bladder urothelial carcinoma (BLCA), breast invasive carcinoma (BRCA), colorectal adenocarcinoma (COADREAD), head and neck squamous cell carcinoma (HNSC) and stomach adenocarcinoma (STAD). Patch embeddings are the released 1,536-dimensional UNI2-h features [4], [44] (256×256 patches at 20× magnification), used as released. The RNA-seq matrices (4,999 genes), clinical metadata and fivefold patient-level splits are those released with SurvPath [2]. The primary endpoint is disease-specific survival (DSS) from the TCGA Clinical Data Resource [45]; overall survival (OS) is secondary. The split is fixed per cohort, so fold k holds the same validation patients for every method and seed; a validation fold has 5–45 events (median 16).

B. Protocol

One protocol is applied without change to every method (Supplementary Table S1). The unit of analysis is the patient, with all slides of a patient concatenated. Every method trains for a fixed budget of 20 epochs and reports the final checkpoint, the default of the MCAT and SurvPath code (no early stopping); no checkpoint is selected on the validation fold, which is also the reported fold. The budget was set after inspecting pooled validation curves under a 10-epoch budget (Supplementary Section S-II), so 10-epoch results are reported as well. Time bins are the quartiles of uncensored training patients, with outer edges $\pm\infty$, reused for validation. The primary metric is Harrell's C-index on continuous time [46],

[47]; Uno's inverse-probability-of-censoring-weighted (IPCW) C-index [48], the integrated Brier score [49] and the log-rank test at the median risk split are secondary (definitions in Supplementary Table S1; results in Sections S-III and S-VI). Three seeds (0, 1, 2) are run, with the fold index added to the seed; Supplementary Section S-VII lists evaluation pitfalls of survival MIL pipelines that the protocol avoids.

Statistics. The three seeds of a fold share its validation patients, so (fold, seed) pairs are not independent replicates [50]. The unit of analysis is therefore the (cohort, fold): the C-index of each method is averaged over the seeds available for both methods, the paired difference is formed on each of the 25 folds, and a paired t-test, a two-sided Wilcoxon signed-rank test (exact; the asymptotic form where paired differences are tied or exactly zero, which occurs only on OS) and a 95% percentile-bootstrap confidence interval (CI) of the mean difference (10,000 resamples of folds) are computed over the folds. Per-cohort tests have five folds each; their p-values are Holm-adjusted over the five cohorts and are exploratory. Pooled tests are not adjusted across comparisons; the Holm family is the five cohorts of one comparison. Tables report the mean and sample standard deviation (sd, $n-1$ denominator) over seeds of the fivefold mean unless the caption says otherwise.

C. Methods

SurvPath (the official model code, trained in our loop with its published optimizer, loss weighting, sampling and patch budget; 21.2 M parameters), ABMIL [8], a self-normalizing network (SNN) [51] on RNA and a two-layer RNA MLP (methods and seeds: Supplementary Table S2) use the hyperparameters of the SurvPath repository; none was tuned under this protocol. During training SurvPath subsamples 4,096 patches per patient from the concatenated slides, its published setting, and reads all patches at evaluation; NullQ-Former reads all patches of a patient in both phases (6,214 for the median of the 40 profiled BLCA patients, Supplementary Table S14). SurvPath's C-index values here are those of its official model code under this protocol with these patch features, not a reproduction of its authors' values; every comparison is within the protocol of Section III-B, and the audit describes the checkpoints trained here, not the published model. Two further baselines are a slide-only and an RNA-only NullQ-Former, trained with the other modality absent, and a late-fusion ensemble that averages their z-scored risks (same seed; two seeds). The two co-primary models are NullQ-Former without auxiliary heads (null codes and modality dropout; 16.9 M parameters) and NullQ-Former (the same plus auxiliary heads, $\lambda = 0.5$), trained with AdamW [52] (SurvPath: RAdam [53], its published setting; Supplementary Table S3), three seeds each.

NullQ-Former. Fig. 1 shows the model; equations and hyperparameters are in Supplementary Section S-II, compute cost in Supplementary Table S14 and the attention over pathway tokens, which is descriptive only [54], in Supplementary Fig. S2. Patient i has patch embeddings $X_i^h \in \mathbb{R}^{N_i \times 1536}$ from all of the patient's slides and a gene vector $x_i^g \in \mathbb{R}^{4999}$, min–max scaled to $[-1, 1]$ on the training split. Genes are grouped into $P = 275$ pathway tokens by per-pathway two-layer MLPs, using the Reactome [55] and MSigDB Hallmark [56] definitions released with SurvPath (331 pathways), of which the 56 with fewer than three member genes among the 4,999 are dropped (the largest pathway has 199 genes); the SurvPath baseline receives the same 275 tokens. Each modality $m \in \{h, g\}$ has $K = 32$ learned queries and two pre-norm blocks of cross-attention to its input tokens, self-attention among the queries and a feed-forward layer, which return a code $Z^m \in \mathbb{R}^{32\times256}$. The two codes are concatenated and passed through two Transformer encoder layers [57]; attention pooling and a sigmoid layer give $T = 4$ discrete-time hazards h_t (bins as in Section III-B), survival $S(t) = \prod_{k\le t}(1 - h_k)$ and risk $r = -\sum_t S(t)$, as in MCAT and SurvPath [58]. For a patient with bin y and censoring indicator c the loss is the negative log-likelihood

$$\mathcal{L}(y, c) = -(1 - c)\big[\log S(y - 1) + \log h_y\big] - c \log S(y), \quad (1)$$

with the censored term unweighted for NullQ-Former and weighted by 0.5 for SurvPath ($\alpha_{\text{surv}} = 0.5$ in its published run scripts). *Training for missingness.* A learned null code $N^m \in \mathbb{R}^{32\times256}$ replaces Z^m when modality m is absent. During training each present modality is dropped with probability 0.15 per sample, independently, with at least one modality kept, and its code is replaced by the null code, so the fusion Transformer sees null codes in training as it will at test time. *Auxiliary heads.* In the full model, attention pooling of Z^h and of Z^g each feeds a linear hazard head trained with (1) on the samples in which that modality is present; the objective is $\mathcal{L}_{\text{fused}} + \lambda \sum_m \mathbf{1}[m \text{ present}]\, \mathcal{L}_m$ with $\lambda = 0.5$, and the heads are discarded at test time. "NullQ-Former without aux. heads" is the same model trained with $\lambda = 0$.

D. The Input-Use Audit

Each trained checkpoint is evaluated on its validation fold under five conditions: *Complete*; *RNA removed*; *RNA permuted*, where the gene vectors are shuffled across the validation patients so that every patient receives a real but wrong profile; *Slide removed*; and *Slide permuted*. For NullQ-Former a removed modality is replaced by its null code. SurvPath cannot run without both inputs, so a removed input is replaced by its training mean: the mean gene vector, or a single-patch bag equal to the mean training patch embedding (over the slides of 60 training patients); NullQ-Former is also evaluated under this mean imputation, to separate the effect of the null code from that of the architecture. Permutation is the stronger test (the perceptual score of Gat et al. [35] applied to survival ranking): a mean or null input is neutral, whereas a permuted input is plausible and wrong, so a model that reads it must lose concordance. In the *random-missingness sweep*, 10, 20, 30 or 50% of validation patients lack one modality at random (five draws per fold, mean reported). Finally, the *risk-score correlation* is the Spearman ρ between a checkpoint's fused risks and the risks of its own RNA-removed and slide-removed evaluations on the same fold, and between its fused risks and those of the other methods on the same fold and

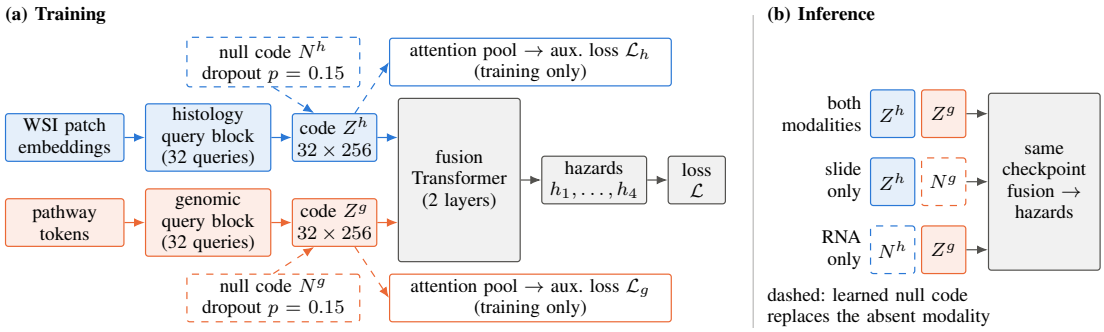


Fig. 1. NullQ-Former is trained for missingness and served from one checkpoint. (a) Training: the WSI patch embeddings and the pathway tokens are each read by 32 learned queries into a code ($Z^h, Z^g \in \mathbb{R}^{32\times256}$); a two-layer fusion Transformer maps the codes to four discrete-time hazards and the survival loss $\mathcal{L}$. Modality dropout ($p = 0.15$ per sample) replaces a present code by its learned null code N^h or N^g (dashed), and auxiliary losses $\mathcal{L}_h$, $\mathcal{L}_g$ on the attention-pooled codes make each branch prognostic on its own (training only). (b) Inference: complete, slide-only and RNA-only patients enter the same checkpoint, with the absent modality's code replaced by its null code.

seed. The variant without auxiliary heads has no permuted evaluation.

IV. Results

A. Do the Models Use Both Inputs?

Table II is the audit, with every p-value, and Fig. 2 plots its pooled differences. SurvPath's C-index is unchanged when its RNA input is removed or permuted across patients (+0.001 in both cases): the model ranks patients the same way with their own profile, the training mean or another patient's profile. Removing the slide costs it 0.073 and permuting it 0.068, leaving 0.47–0.52 (removed) and 0.46–0.53 (permuted) on every cohort; the smallest change under removal is on BRCA (−0.016; −0.004 under permutation), where its complete-input score (0.536) is already close to chance. NullQ-Former without auxiliary heads behaves the same way with respect to RNA (RNA removed +0.000) but loses only 0.030 without the slide. With auxiliary heads the fused prediction responds to RNA: −0.013 when it is removed and −0.023 when it is permuted ($p = 0.075$ and 0.063 over the 25 seed-averaged folds, short of 0.05; the attribution rests on this direction and on the contrast with the +0.000 and +0.001 of the other two models), the ordering expected of an input that is read; removing the slide costs 0.025 and permuting it 0.067, and the C-index stays at 0.54–0.62 with either input removed.

The risk-score correlations say how the fused prediction is composed. For NullQ-Former, the Spearman ρ between the fused risk and the same checkpoint's RNA-removed risk is 0.83 pooled over folds and seeds, against 0.45 with its slide-removed risk, and the two single-input risks correlate at 0.15 (Supplementary Table S9, which also gives its correlations with the other models' risks). The fused prediction therefore remains histology-led: the auxiliary heads make it respond to RNA, they do not make the two inputs symmetric.

B. Accuracy at Equal Budget

Table III gives the C-index of every method and Table IV the seed-averaged paired tests. Pooled over 25 folds, NullQ-Former exceeds SurvPath by 0.036 (Wilcoxon $p = 0.030$, t-test $p = 0.074$) and the variant without auxiliary heads by 0.037 ($p = 0.020$ and 0.020); Table IV gives the CIs

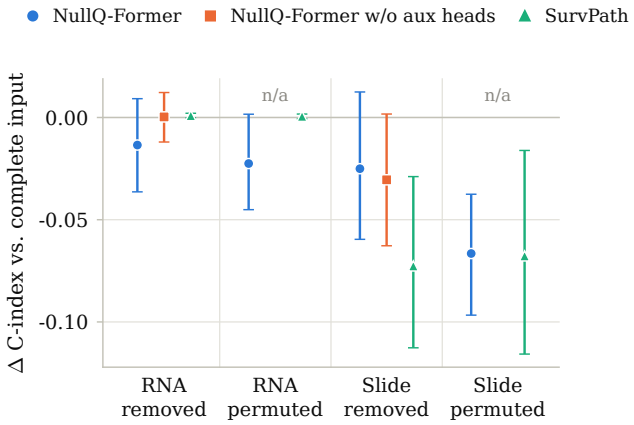


Fig. 2. Change in C-index of the same 20-epoch checkpoint when one test-time input is removed or permuted across validation patients, relative to its complete-input evaluation, pooled over the five cohorts: mean and 95% bootstrap CI over the 25 seed-averaged folds (10,000 resamples). Removed inputs as in Table II, which gives the per-cohort values; "w/o aux heads" was not evaluated under permutation.

and folds won. Per cohort, only BRCA has a raw $p < 0.05$ (+0.086, t-test $p = 0.017$; Holm-adjusted over the five cohorts $p = 0.084$ under the t-test and 0.312 under Wilcoxon), so no DSS cohort survives correction (Supplementary Table S5). Against the RNA-only MLP the margins are +0.012 and +0.014 ($p \geq 0.33$), and against the late-fusion ensemble +0.011 and +0.009 ($p \geq 0.27$; seeds 0–1); the MLP scores higher than NullQ-Former on BRCA and COADREAD and higher than the variant without auxiliary heads on BLCA and BRCA. No unimodal baseline differs from SurvPath pooled (ABMIL +0.001, SNN −0.007, MLP +0.024). The equal-budget result is therefore parity with the best unimodal baseline, not superiority. On overall survival (one seed) NullQ-Former leads SurvPath by 0.034 ($p = 0.003$ and 0.0025), and under a 10-epoch budget the variant without auxiliary heads leads by 0.028 ($p = 0.079$ and 0.063); both are in Table IV and Supplementary Tables S6 and S7. Uno's IPCW C-index tracks Harrell's: pooled over the 25 seed-averaged folds it exceeds SurvPath's by 0.039 for NullQ-Former and 0.046 for the variant without auxiliary heads (Wilcoxon $p = 0.042$ for both; t-test $p = 0.153$ and 0.045). The integrated Brier score of both fusion variants, ABMIL and the RNA MLP is about

TABLE II

INPUT-USE AUDIT: C-INDEX OF THE SAME 20-EPOCH CHECKPOINT UNDER THE FOUR ALTERED TEST-TIME CONDITIONS OF SECTION III-D, MEAN ± SD OVER THREE SEEDS OF THE FIVEFOLD MEAN; THE COMPLETE CONDITION IS THE WSI+RNA ROW OF TABLE III. REMOVED INPUT: TRAINING MEAN FOR SURVPATH, LEARNED NULL CODE FOR NULLQ-FORMER; PERMUTED INPUT: SHUFFLED ACROSS THE VALIDATION PATIENTS; THE VARIANT WITHOUT AUXILIARY HEADS WAS NOT EVALUATED UNDER PERMUTATION. Δ: POOLED DIFFERENCE FROM COMPLETE OVER THE 25 SEED-AVERAGED FOLDS, WITH PAIRED t-TEST AND WILCOXON p-VALUES.

Condition	BLCA[†]	BRCA	COADREAD	HNSC	STAD	Δ vs. Complete	t-test p	Wilcoxon p
SurvPath (official), mean imputation								
RNA removed	0.595 ± 0.013	0.536 ± 0.027	0.572 ± 0.037	0.553 ± 0.018	0.588 ± 0.007	$+0.001$	0.12	0.085
RNA permuted	0.594 ± 0.013	0.537 ± 0.028	0.572 ± 0.038	0.552 ± 0.011	0.587 ± 0.006	$+0.001$	0.36	0.13
Slide removed	0.486 ± 0.022	0.520 ± 0.025	0.473 ± 0.012	0.504 ± 0.043	0.493 ± 0.082	-0.073	0.003	0.002
Slide permuted	0.511 ± 0.018	0.532 ± 0.052	0.510 ± 0.024	0.489 ± 0.033	0.459 ± 0.016	-0.068	0.016	0.007
NullQ-Former without aux. heads, null codes								
RNA removed	0.612 ± 0.020	0.600 ± 0.046	0.633 ± 0.027	0.588 ± 0.008	0.595 ± 0.021	$+0.000$	0.96	0.51
Slide removed	0.590 ± 0.050	0.572 ± 0.033	0.601 ± 0.056	0.533 ± 0.017	0.578 ± 0.035	-0.030	0.080	0.16
NullQ-Former, null codes								
RNA removed	0.600 ± 0.002	0.592 ± 0.047	0.576 ± 0.025	0.596 ± 0.016	0.589 ± 0.010	-0.013	0.27	0.43
RNA permuted	0.597 ± 0.010	0.572 ± 0.046	0.618 ± 0.028	0.566 ± 0.019	0.555 ± 0.007	-0.023	0.075	0.063
Slide removed	0.613 ± 0.022	0.617 ± 0.052	0.583 ± 0.022	0.545 ± 0.026	0.538 ± 0.004	-0.025	0.19	0.11
Slide permuted	0.570 ± 0.035	0.539 ± 0.027	0.556 ± 0.033	0.539 ± 0.058	0.484 ± 0.035	-0.067	0.0002	0.0004

[†]Development cohort.

0.07 lower than SurvPath's ($p < 0.001$), whose published loss weights the censored term by 0.5 and leaves its hazards poorly calibrated under this protocol, so the Brier comparison is reported for completeness only (Supplementary Table S8).

C. Missing Inputs at Test Time

Fig. 3 shows the random-missingness sweep. With half of the validation patients missing one modality, the fused C-index of NullQ-Former changes by at most 0.025 (BRCA, 0.622 to 0.597) and that of the variant without auxiliary heads by at most 0.018 (BRCA). For NullQ-Former every change is within one seed standard deviation of the complete-input score (Table III); for the variant without auxiliary heads the HNSC and STAD changes (0.013 and 0.017) slightly exceed their seed standard deviations (0.012 and 0.015). A site with mixed availability can instead route slide-only patients to ABMIL and RNA-only patients to the RNA MLP, or average the two unimodal NullQ-Formers when both inputs exist; routing and late fusion are robust by construction, and late fusion matches the joint model (Table IV). What the single checkpoint offers over routing is operational: one model to validate, version and monitor, and one output head whose risks are produced the same way for every patient (whether they are comparable across missingness patterns was not tested). The result is about coverage, not accuracy: with the RNA removed, the single NullQ-Former checkpoint scores 0.576–0.600 against 0.553–0.587 for ABMIL (higher on four cohorts, lower on COADREAD); with the slide removed, 0.538–0.617 against 0.532–0.633 for the RNA MLP (higher on BLCA and STAD, lower on BRCA, COADREAD and HNSC, by up to 0.05 on COADREAD; three-seed means).

D. Where the Robustness Comes From

Evaluating the same checkpoints under mean imputation instead of the null code separates the two. For the variant without auxiliary heads, replacing the removed slide by its mean imputation instead of the null code turns a loss of 0.030 (0.026 at seed 0) into a loss of 0.069 (t-test $p = 3 \times 10^{-4}$, Wilcoxon $p = 2.5 \times 10^{-4}$; one seed, 25 folds), the size of SurvPath's collapse; with the heads the loss is 0.035 under mean imputation ($p = 0.040$ and 0.045; three seeds) against 0.025 with the null code (per-cohort values in Supplementary Table S10). The attribution is therefore: null codes and modality dropout remove the collapse without the slide; the auxiliary heads make the fused output respond to RNA; and the fused risk remains histology-led (ρ 0.83 against 0.45). Fusion-block ablations on BLCA under the 10-epoch budget (one seed) move the fused C-index within 0.595–0.638, inside the fold standard deviation of 0.07–0.13 (Supplementary Table S12). Under the 20-epoch budget on BLCA (one seed, CPU), capping NullQ-Former at SurvPath's 4,096 training patches per patient gives 0.626 against 0.625 for the seed-0 run with all patches, so on this cohort the margin over SurvPath is not explained by NullQ-Former reading more patches; two or eight time bins, SurvPath's Xena pathway set [59] (225 tokens after the three-gene rule) and the 50 Hallmark gene sets alone stay within one fold standard deviation (0.600–0.628; Supplementary Table S13).

E. Checkpoint Selection on Event-Poor Folds

Every run stores its per-epoch validation history, so Supplementary Table S11 replays validation-loss early stopping (patience 5) on the 590 stored fold histories of the eight 20-epoch methods without retraining. The rule selects epoch 1 on median (mean 1.6; 93% of folds epoch 1–3) and would have stopped after 6.6 of 20 epochs on average, with the patience limit reached in every fold. Pooled over the 25 seed-averaged folds, the early-stopped checkpoint scores below the final one by 0.090 for NullQ-Former without heads, 0.059 for NullQ-Former, 0.065 for the slide-only and 0.034 for the RNA-only NullQ-Former, 0.038 for SNN and 0.027 for the RNA MLP, and by nothing for ABMIL (-0.001) and SurvPath ($+0.000$). The histories show why: ABMIL and SurvPath reach their best validation C-index at epoch 6.2 and 6.9 on average, SNN and the MLP at 7.2–7.3, and the NullQ-Former variants, which train with a warm-up epoch and a lower learning rate (Supplementary Table S3), at 9.3–10.0, so an epoch-1–2 checkpoint costs them most. Conversely, under the oracle rule (the epoch

TABLE III
DISEASE-SPECIFIC SURVIVAL C-INDEX UNDER THE 20-EPOCH PROTOCOL: MEAN ± SD OVER SEEDS OF THE FIVEFOLD MEAN (THREE SEEDS; LATE FUSION TWO; THE BRCA CELLS OF THE SINGLE-MODALITY NULLQ-FORMERS TWO). POOLED: MEAN OF THE FIVE COHORT MEANS.

Method	Inputs	BLCA[†]	BRCA	COADREAD	HNSC	STAD	Pooled
NullQ-Former	WSI+RNA	0.630 ± 0.010	0.622 ± 0.033	0.613 ± 0.023	0.582 ± 0.025	0.572 ± 0.022	0.604
NullQ-Former without aux. heads	WSI+RNA	0.609 ± 0.016	0.611 ± 0.039	0.638 ± 0.031	0.575 ± 0.012	0.593 ± 0.015	0.605
SurvPath	WSI+RNA	0.594 ± 0.012	0.536 ± 0.026	0.570 ± 0.038	0.552 ± 0.019	0.587 ± 0.008	0.568
Late fusion	WSI+RNA	0.629 ± 0.018	0.606 ± 0.020	0.620 ± 0.069	0.571 ± 0.009	0.544 ± 0.006	0.594
ABMIL	WSI	0.566 ± 0.011	0.575 ± 0.004	0.587 ± 0.023	0.562 ± 0.028	0.553 ± 0.020	0.569
Slide-only NullQ-Former	WSI	0.592 ± 0.003	0.615 ± 0.027	0.574 ± 0.024	0.585 ± 0.003	0.582 ± 0.019	0.590
SNN	RNA	0.590 ± 0.009	0.558 ± 0.013	0.572 ± 0.028	0.535 ± 0.006	0.549 ± 0.009	0.561
RNA MLP	RNA	0.611 ± 0.012	0.627 ± 0.013	0.633 ± 0.016	0.555 ± 0.012	0.532 ± 0.009	0.592
RNA-only NullQ-Former	RNA	0.608 ± 0.020	0.559 ± 0.001	0.598 ± 0.035	0.554 ± 0.013	0.511 ± 0.005	0.566

[†]Development cohort.

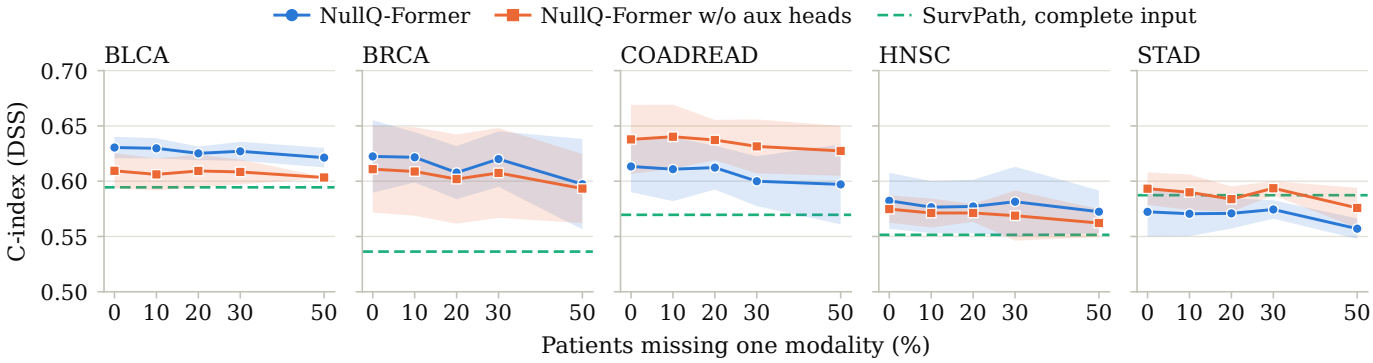


Fig. 3. C-index of NullQ-Former and of its variant without auxiliary heads (20 epochs) when 0–50% of validation patients are missing one modality at random (mean of five draws per fold). Lines: mean over three seeds of the fivefold mean; bands: ±1 sd over seeds; dashed: SurvPath with complete inputs (three-seed mean).

TABLE IV
SEED-AVERAGED PAIRED TESTS OVER THE 25 POOLED (COHORT, FOLD) UNITS (DSS AT 20 EPOCHS UNLESS MARKED OS OR 10 EP.; THREE SEEDS FOR DSS, SEED 0 FOR OS; SEEDS AVERAGED PER FOLD BEFORE PAIRING): MEAN Δ C-INDEX (FIRST MINUS SECOND METHOD), 95% PERCENTILE-BOOTSTRAP CI (10,000 FOLD RESAMPLES), PAIRED t-TEST AND WILCOXON p-VALUES, FOLDS WON. NQF: NULLQ-FORMER; NQF$^-$: ITS VARIANT WITHOUT AUXILIARY HEADS. UNIMODAL BASELINES VS. SURVPATH: SUPPLEMENTARY TABLE S4.

Comparison	Δ 95% CI	t-test p	Wilcoxon p	Wins
NQF vs. SurvPath	$+0.036$ $[-0.004, +0.071]$	0.074	0.030	18/25
NQF$^-$ vs. SurvPath	$+0.037$ $[+0.009, +0.065]$	0.020	0.020	17/25
NQF vs. RNA MLP	$+0.012$ $[-0.022, +0.044]$	0.47	0.34	14/25
NQF$^-$ vs. RNA MLP	$+0.014$ $[-0.023, +0.049]$	0.47	0.37	16/25
NQF vs. late fusion[a]	$+0.011$ $[-0.013, +0.034]$	0.39	0.28	14/25
NQF$^-$ vs. late fusion[a]	$+0.009$ $[-0.020, +0.036]$	0.54	0.52	14/25
NQF vs. SurvPath (OS)	$+0.034$ $[+0.014, +0.054]$	0.003	0.0025	19/25
NQF$^-$ vs. SurvPath (10 ep.)	$+0.028$ $[-0.001, +0.057]$	0.079	0.063	18/25

[a]Seeds 0–1 (late fusion was run with two seeds).

of highest validation C-index, an unattainable bound biased in favor of every method) the pooled C-index is 0.67–0.68 for SurvPath, ABMIL, SNN and the RNA MLP and 0.69 for both NullQ-Former variants (Supplementary Table S11), so the margin in Table IV depends on the fixed-budget, final-checkpoint rule as well as on the models. Under the replayed rule 23 of the 40 method × cohort cells have a mean C-index within 0.45–0.55, against 5 under the final checkpoint (HNSC 8 of 8, STAD 7 of 8, BRCA 4 of 8); an earlier campaign that actually trained with validation-loss selection behaved the same way (median selected epoch 1; 14 of 25 cells within 0.45–0.55; Supplementary Section S-VIII). Every replayed rule selects and scores on the same fold, so the measured deficit of early stopping is conservative. On cohorts with 5–45 events per validation fold, checkpoints should not be selected on the validation negative log-likelihood; a fixed budget with the final checkpoint, or at least the selected epochs and the events per fold, should be reported.

V. DISCUSSION

A general lesson. Fusion of a strong and a weak modality can converge to ignoring the weak one with no accuracy signal: the fused C-index of the SurvPath checkpoints trained here is as high as their slide-only behavior permits, and nothing in the survival objective penalizes leaving the genomic branch unused once the fusion block has found a low-loss route through the histology branch. This is modality competition [41], which Gradient Blending [42] and OGM-GE [43] counteract in classification, and the same remedy applies: per-modality supervision supplies a direct gradient to the weak branch. The audit detects the failure without any training: removal and permutation across patients, with the risk-score correlation, show which input a ranking responds to, and they should accompany any multimodal C-index. Unimodal supervision makes the fused output respond to RNA ($p = 0.075$ and 0.063), but the fused risk remains histology-led (ρ 0.83 against 0.45, Section IV-A) and the RNA-permuted loss (0.023) is a third of the slide-permuted loss (0.067).

The RNA-only baseline. A two-layer MLP on bulk expression, run with repository defaults, matches both fusion variants pooled and scores higher than NullQ-Former on two of five cohorts. This cheap baseline is seldom reported at equal

budget; without it, an improvement attributed to fusion may belong to the genomic branch. Separating "pathway tokens help" from "fusion helps" would need an MLP that receives the same pathway tokens, which we have not run.

Limitations. **Baseline tuning and variant selection:** SurvPath was not re-tuned and NullQ-Former's hyperparameters and variants were chosen as described under Scope of claims and in Supplementary Section S-II, so the accuracy comparison is favorable to NullQ-Former by an amount that cannot be bounded without tuning the baseline; its margin shrinks to about 0.01 under oracle selection (Section IV-E).

Recipe on other architectures: null codes, modality dropout and auxiliary heads have not been applied to SurvPath, so how much of the robustness is specific to the query bottleneck is untested.

Seeds and statistics: three seeds for the main comparison, two for late fusion, one for OS. Per-cohort tests have five folds; the fold-level standard deviation of the C-index is 0.12–0.25 on COADREAD for the fusion models (0.05–0.25 over all methods), fold 3 of BLCA is weak for every method except the RNA-only NullQ-Former, and fold 0 of COADREAD, with five events, scores 0.85–0.92 for NullQ-Former across seeds. No per-cohort DSS difference survives correction; on OS (one seed) BLCA, the development cohort, does under the t-test only (Holm-adjusted $p = 0.039$ under the t-test and 0.312 under Wilcoxon). The RNA-permuted loss of 0.023 that supports the auxiliary-head claim has $p = 0.075$ and 0.063 over 25 folds.

Data and clinical use: TCGA only, with frozen UNI2-h features whose pretraining corpus may include TCGA slides, which would make every WSI-based result optimistic; no external cohort; missingness simulated at random, so informative missingness is not covered; no clinical covariates (stage, grade, age), so whether any model adds to them is untested; performance across sex, race and age subgroups was not evaluated. The model is a research prototype evaluated on public, de-identified data, not for patient-level decision support, and it must not be used to inform patient care.

VI. Conclusion

A fused histology–transcriptomics survival model can score well while its ranking ignores one input: under an equal-budget protocol on five TCGA cohorts, the official SurvPath implementation does not respond to its RNA input and collapses without the slide, which removal and permutation audits with risk-score correlations detect. Null codes and modality dropout remove the collapse and auxiliary unimodal heads make the fused prediction respond to RNA (a permutation loss of 0.023, $p = 0.075$ and 0.063); the resulting single checkpoint serves complete, slide-only and RNA-only patients, is not detectably different from an RNA-only MLP or a late-fusion ensemble with both inputs, and stays within about 0.05 of the dedicated unimodal baseline with one input absent. Replaying checkpoint selection on the stored histories shows that validation-loss early stopping would have selected epoch 1–3 checkpoints and penalized the late-learning models on these event-poor folds, which argues for fixed budgets and for reporting selected epochs and events per fold.

Data and Code Availability

TCGA data are public and de-identified. UNI2-h patch features [44] and the RNA-seq matrices, clinical metadata and splits released with SurvPath [2] are available from their authors. The code (MIT license), configurations, archived per-run results and the table-regeneration script are public at github.com/SharoonSharif/PathQFormer (the repository, archive and checkpoints carry the model's earlier name, PathQ-Former) and archived at Zenodo (version 0.3.0, doi:10.5281/zenodo.23245611). The 275 fold models (Supplementary Section S-VIII; 18 GB) are at huggingface.co/sharoonsharif1/PathQFormer-checkpoints. One supplementary PDF (Sections S-I–S-VIII) accompanies the paper.

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References

[1] R. J. Chen *et al.*, "Pan-cancer integrative histology-genomic analysis via multimodal deep learning," *Cancer Cell*, vol. 40, no. 8, pp. 865–878.e6, 2022.

[2] G. Jaume, A. Vaidya, R. J. Chen, D. F. K. Williamson, P. P. Liang, and F. Mahmood, "Modeling dense multimodal interactions between biological pathways and histology for survival prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 11 579–11 590.

[3] A. H. Song, R. J. Chen, G. Jaume, A. J. Vaidya, A. S. Baras, and F. Mahmood, "Multimodal prototyping for cancer survival prediction," in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, vol. 235, 2024, pp. 46 050–46 073.

[4] R. J. Chen *et al.*, "Towards a general-purpose foundation model for computational pathology," *Nat. Med.*, vol. 30, pp. 850–862, 2024.

[5] H. Xu *et al.*, "A whole-slide foundation model for digital pathology from real-world data," *Nature*, vol. 630, pp. 181–188, 2024.

[6] E. Vorontsov *et al.*, "A foundation model for clinical-grade computational pathology and rare cancers detection," *Nat. Med.*, vol. 30, pp. 2924–2935, 2024.

[7] R. J. Chen *et al.*, "Multimodal co-attention transformer for survival prediction in gigapixel whole slide images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 4015–4025.

[8] M. Ilse, J. Tomczak, and M. Welling, "Attention-based deep multiple instance learning," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, vol. 80, 2018, pp. 2127–2136.

[9] M. Y. Lu, D. F. K. Williamson, T. Y. Chen, R. J. Chen, M. Barbieri, and F. Mahmood, "Data-efficient and weakly supervised computational pathology on whole-slide images," *Nat. Biomed. Eng.*, vol. 5, no. 6, pp. 555–570, 2021.

[10] Z. Shao *et al.*, "TransMIL: Transformer based correlated multiple instance learning for whole slide image classification," in *Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 2136–2147.

[11] X. Wang *et al.*, "Transformer-based unsupervised contrastive learning for histopathological image classification," *Med. Image Anal.*, vol. 81, p. 102559, 2022.

[12] P. Mobadersany *et al.*, "Predicting cancer outcomes from histology and genomics using convolutional networks," *Proc. Natl. Acad. Sci. USA*, vol. 115, no. 13, pp. E2970–E2979, 2018.

[13] R. J. Chen *et al.*, "Pathomic fusion: An integrated framework for fusing histopathology and genomic features for cancer diagnosis and prognosis," *IEEE Trans. Med. Imag.*, vol. 41, no. 4, pp. 757–770, 2022.

[14] Y. Xu and H. Chen, "Multimodal optimal transport-based co-attention transformer with global structure consistency for survival prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 21 241–21 251.

[15] F. Zhou and H. Chen, "Cross-modal translation and alignment for survival analysis," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 21 485–21 494.

[16] Y. Zhang, Y. Xu, J. Chen, F. Xie, and H. Chen, "Prototypical information bottlenecking and disentangling for multimodal cancer survival prediction," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[17] J. Peng *et al.*, "Continually evolved multimodal foundation models for cancer prognosis," 2025, arXiv:2501.18170.

[18] H. Liu, H. Yang, F. Eduati, J. P. W. Pluim, and M. Veta, "Adaptive prototype learning for multimodal cancer survival analysis," 2025, arXiv:2503.04643.

[19] N. Neverova, C. Wolf, G. Taylor, and F. Nebout, "ModDrop: Adaptive multi-modal gesture recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 8, pp. 1692–1706, 2016.

[20] A. Cheerla and O. Gevaert, "Deep learning with multimodal representation for pancancer prognosis prediction," *Bioinformatics*, vol. 35, no. 14, pp. i446–i454, 2019.

[21] L. A. Vale-Silva and K. Rohr, "Long-term cancer survival prediction using multimodal deep learning," *Sci. Rep.*, vol. 11, p. 13505, 2021.

[22] M. Ma, J. Ren, L. Zhao, D. Testuggine, and X. Peng, "Are multimodal transformers robust to missing modality?" in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 18 177–18 186.

[23] M. Ma, J. Ren, L. Zhao, S. Tulyakov, C. Wu, and X. Peng, "SMIL: Multimodal learning with severely missing modality," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2021, pp. 2302–2310.

[24] H. Wang, Y. Chen, C. Ma, J. Avery, L. Hull, and G. Carneiro, "Multi-modal learning with missing modality via shared-specific feature modelling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 15 878–15 887.

[25] Y.-L. Lee, Y.-H. Tsai, W.-C. Chiu, and C.-Y. Lee, "Multimodal prompting with missing modalities for visual recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 14 943–14 952.

[26] C. Cui *et al.*, "Survival prediction of brain cancer with incomplete radiology, pathology, genomic, and demographic data," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, vol. 13435, 2022, pp. 626–635.

[27] K. Ding, M. Zhou, D. N. Metaxas, and S. Zhang, "Pathology-and-genomics multimodal transformer for survival outcome prediction," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, vol. 14225, 2023, pp. 622–631.

[28] Z. Wang, Y. Zhang, Y. Xu, S. Imoto, H. Chen, and J. Song, "Histo-genomic knowledge association for cancer prognosis from histopathology whole slide images," *IEEE Trans. Med. Imag.*, vol. 44, no. 5, pp. 2170–2181, 2025.

[29] L. Qu, D. Huang, S. Zhang, and X. Wang, "Multi-modal data binding for survival analysis modeling with incomplete data and annotations," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, vol. 15005, 2024, pp. 501–510.

[30] Y. Xu, F. Zhou, C. Zhao, Y. Wang, C. Yang, and H. Chen, "Distilled prompt learning for incomplete multimodal survival prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 5102–5111.

[31] J. Zhou, J. Tang, Y. Zuo, P. Wan, D. Zhang, and W. Shao, "Robust multimodal survival prediction with conditional latent differentiation variational autoencoder," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 10 384–10 393.

[32] F. Liu *et al.*, "Prototype-guided cross-modal knowledge enhancement for adaptive survival prediction," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, vol. 15965, 2025, pp. 522–532.

[33] M. Qu *et al.*, "Memory-augmented incomplete multimodal survival prediction via cross-slide and gene-attentive hypergraph learning," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, vol. 15969, 2025, pp. 318–327.

[34] K. Yu *et al.*, "PRIME: Prototype-driven multimodal pretraining for cancer prognosis with missing modalities," 2026, arXiv:2604.04999.

[35] I. Gat, I. Schwartz, and A. Schwing, "Perceptual score: What data modalities does your model perceive?" in *Adv. Neural Inf. Process. Syst.*, vol. 34, 2021.

[36] J. Lee, Y. Lee, J. Kim, A. Kosiorek, S. Choi, and Y. W. Teh, "Set transformer: A framework for attention-based permutation-invariant neural networks," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, vol. 97, 2019, pp. 3744–3753.

[37] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, vol. 12346, 2020, pp. 213–229.

[38] A. Jaegle, F. Gimeno, A. Brock, O. Vinyals, A. Zisserman, and J. Carreira, "Perceiver: General perception with iterative attention," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, vol. 139, 2021, pp. 4651–4664.

[39] J.-B. Alayrac *et al.*, "Flamingo: a visual language model for few-shot learning," in *Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 23 716–23 736.

[40] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, vol. 202, 2023, pp. 19 730–19 742.

[41] Y. Huang, J. Lin, C. Zhou, H. Yang, and L. Huang, "Modality competition: What makes joint training of multi-modal network fail in deep learning? (Provably)," in *Proc. 39th Int. Conf. Mach. Learn. (ICML)*, vol. 162, 2022, pp. 9226–9259.

[42] W. Wang, D. Tran, and M. Feiszli, "What makes training multi-modal classification networks hard?" in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 12 692–12 702.

[43] X. Peng, Y. Wei, A. Deng, D. Wang, and D. Hu, "Balanced multimodal learning via on-the-fly gradient modulation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 8238–8247.

[44] Mahmood Lab, "UNI2-h pathology foundation model and precomputed UNI2-h patch features for TCGA, CPTAC and PANDA whole-slide images," Hugging Face model and dataset repositories, 2025, license CC BY-NC-ND 4.0; accessed Sep. 2026; features at https://huggingface.co/datasets/MahmoodLab/UNI2-h-features. [Online]. Available: https://huggingface.co/MahmoodLab/UNI2-h

[45] J. Liu *et al.*, "An integrated TCGA pan-cancer clinical data resource to drive high-quality survival outcome analytics," *Cell*, vol. 173, no. 2, pp. 400–416.e11, 2018.

[46] F. E. Harrell, R. M. Califf, D. B. Pryor, K. L. Lee, and R. A. Rosati, "Evaluating the yield of medical tests," *JAMA*, vol. 247, no. 18, pp. 2543–2546, 1982.

[47] S. Pölsterl, "scikit-survival: A library for time-to-event analysis built on top of scikit-learn," *J. Mach. Learn. Res.*, vol. 21, no. 212, pp. 1–6, 2020.

[48] H. Uno, T. Cai, M. J. Pencina, R. B. D'Agostino, and L. J. Wei, "On the C-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data," *Stat. Med.*, vol. 30, no. 10, pp. 1105–1117, 2011.

[49] E. Graf, C. Schmoor, W. Sauerbrei, and M. Schumacher, "Assessment and comparison of prognostic classification schemes for survival data," *Stat. Med.*, vol. 18, no. 17–18, pp. 2529–2545, 1999.

[50] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Mach. Learn.*, vol. 52, no. 3, pp. 239–281, 2003.

[51] G. Klambauer, T. Unterthiner, A. Mayr, and S. Hochreiter, "Self-normalizing neural networks," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 971–980.

[52] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.

[53] L. Liu *et al.*, "On the variance of the adaptive learning rate and beyond," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2020.

[54] S. Jain and B. C. Wallace, "Attention is not explanation," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol. (NAACL-HLT)*, 2019, pp. 3543–3556.

[55] M. Gillespie *et al.*, "The Reactome pathway knowledgebase 2022," *Nucleic Acids Res.*, vol. 50, no. D1, pp. D687–D692, 2022.

[56] A. Liberzon, C. Birger, H. Thorvaldsdóttir, M. Ghandi, J. P. Mesirov, and P. Tamayo, "The molecular signatures database (MSigDB) hallmark gene set collection," *Cell Syst.*, vol. 1, no. 6, pp. 417–425, 2015.

[57] A. Vaswani *et al.*, "Attention is all you need," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.

[58] S. G. Zadeh and M. Schmid, "Bias in cross-entropy-based training of deep survival networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 9, pp. 3126–3137, 2021.

[59] M. J. Goldman *et al.*, "Visualizing and interpreting cancer genomics data via the Xena platform," *Nat. Biotechnol.*, vol. 38, no. 6, pp. 675–678, 2020.